

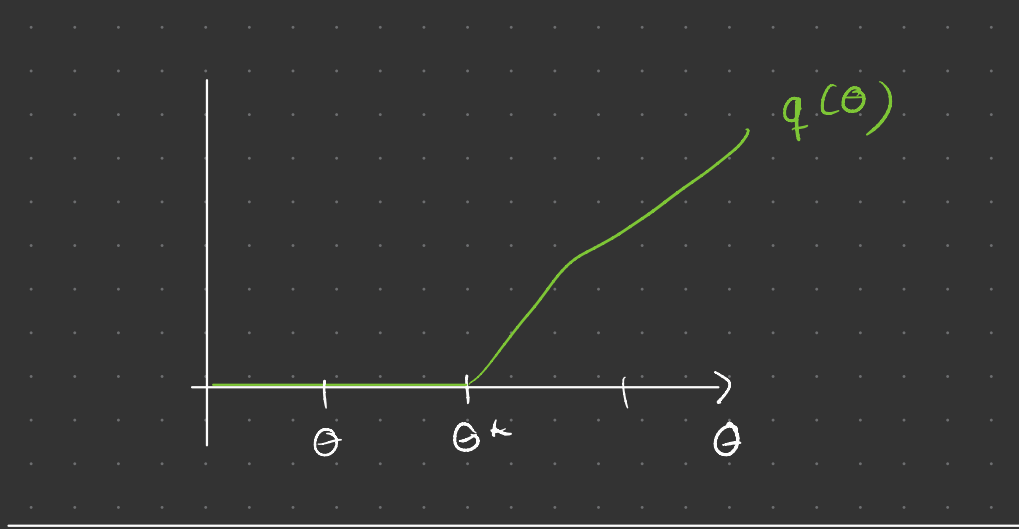

$$t(\theta) = \theta q(\theta) - U(\theta)$$

$$q(\theta) = \max \left\{ \theta - \frac{1-F(\theta)}{f(\theta)}, 0 \right\} = \begin{cases} 0 & \text{if } \theta < \theta^* \\ \theta - \frac{1-F(\theta)}{f(\theta)} & \theta \geq \theta^* \end{cases}$$

θ^* as the solution $\theta^* - \frac{1-F(\theta^*)}{f(\theta^*)}$

$$U(\theta) = \int_{\theta_L}^{\theta} q(\tau) d\tau = \int_{\theta_L}^{\theta} \max \left\{ \tau - \frac{1-F(\tau)}{f(\tau)}, 0 \right\} d\tau$$

$$= \begin{cases} \int_{\theta_L}^{\theta} 0 d\tau = 0 & \text{for } \theta < \theta^* \\ \int_{\theta_L}^{\theta^*} 0 d\tau + \int_{\theta^*}^{\theta} \left(\tau - \frac{1-F(\tau)}{f(\tau)} \right) d\tau & \theta > \theta^* \end{cases}$$



Suppose $\theta \sim U[\theta_L, \theta_H]$

pdf: $f(\theta) = \frac{1}{\theta_H - \theta_L}$, cdf: $F(\theta) = \frac{\theta - \theta_L}{\theta_H - \theta_L}$

$$\frac{1 - F(\theta)}{f(\theta)} = \frac{1 - \frac{\theta - \theta_L}{\theta_H - \theta_L}}{\frac{1}{\theta_H - \theta_L}} = \frac{\theta_H - \cancel{\theta_L} - \theta + \cancel{\theta_L}}{\frac{1}{\theta_H - \cancel{\theta_L}}} = \theta_H - \theta$$

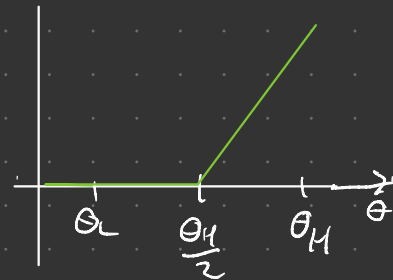
$$q(\theta) = \max\{\theta - (\theta_H - \theta), 0\}$$

$$= \max\{2\theta - \theta_H, 0\}$$

$$\theta^* \text{ solution: } 2\theta^* - \theta_H = 0$$

$$\Rightarrow \theta^* = \frac{\theta_H}{2}$$

$$q(\theta) = \begin{cases} 0 & \text{if } \theta < \frac{\theta_H}{2} \\ 2\theta - \theta_H & \text{if } \theta \geq \frac{\theta_H}{2} \end{cases}$$



$$U(\theta) = \int_{\theta_L}^{\theta} q(\tau) d\tau$$

$$= \int_{\theta_L}^{\theta} \max \{2\tau - \theta_H, 0\} d\tau$$

$$= \begin{cases} 0 & \text{if } \theta < \theta_H/2 \\ \int_{\theta_H/2}^{\theta} (2\tau - \theta_H) d\tau = \left(2 \frac{\tau^2}{2} - \theta_H \tau \right) \Big|_{\theta_H/2}^{\theta} & \theta \geq \frac{\theta_H}{2} \end{cases}$$

$$= (\theta^2 - \theta_H \theta) - \left(\frac{\theta_H^2}{4} - \frac{\theta_H^2}{2} \right)$$

Recall $U(q, t, \theta) = v(q, \theta) - t$

For this problem

$$v(q, \theta) = \theta q$$

$$t(\theta) = \underbrace{\theta q(\theta)}_{= v(q(\theta), \theta)} - U(\theta)$$

$$= v(q(\theta), \theta)$$

For a twice continuously differentiable function $v: \mathbb{R}^2 \rightarrow \mathbb{R}$

$v_{12} = v_{21}$ i.e., cross-partial derivatives are equal.

Ex: $v(q, \theta) = \theta q$

$$v_1 = \theta \quad v_{12} = 1$$

$$v_2 = q \quad v_{21} = 1$$